NEWTONSCHOOLOFTECHNOLOGY Data Visualisation Analytics — Capstone 2

Decoding Beijing's Air Crisis A Data-Driven Analysis of PM2.5 Pollution Patterns (2010–2014)

Sector	Environmental Analytics / Public Health
Team ID	SectionD – G12
Section	D
Faculty Mentor	Archit Raj
Institute	Newton School of Technology
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Team Members

Milind Bansal	Project Lead + Data Lead / ETL Lead	MilindDevX
Vishuti Jamwal	Analysis & Visualisation Lead	Vishuti-Jamwal
Hrishabh	Visualization Lead	hrishu802
Om	Presentation Design	yadavom7345
Samay Samrat	Reasearch & report lead	samay-hash
Praveen Nitharwal	Final Load & Dashboard Lead	Prav1104

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Executive Summary

Project Core: This analysis identifies that Beijing's PM2.5 levels exceeded WHO safe limits by 6–7$\times$ annually (2010–2014). We establish wind speed and seasonal coal heating as the primary drivers of hazardous air quality.

Problem

Fine particulate matter (PM2.5) represents the single greatest environmental health risk in urban China. During 2010–2014, Beijing's air quality remained chronically hazardous without a structured, data-backed decision framework for pollution alerts.

Approach

We executed an end-to-end analytics pipeline on 43,824 hourly observations from the UCI Beijing PM2.5 dataset. This included automated ETL cleaning, 11 business-focused EDA visualisations, 7 non-parametric statistical tests, and a dual-view Tableau dashboard.

Key Insights

- **WHO Breach:** Annual means (89.9–101.6 μg/m^3) were 6–7× higher than the 15 μg/m^3 guideline.
- **Seasonality:** Winter is the most polluted season (109.9 μg/m^3) due to heating emissions.
- **Wind Factor:** Calm wind conditions (cv) resulted in the highest pollution (124.5 μg/m^3), while NW winds act as a natural purifier (69.8 μg/m^3).

Key Recommendations

- Launch winter-priority emission reduction programmes targeting coal heating.
- Implement "Clean Window" alerts triggered by North-Westerly wind forecasts.

- Deploy calm-wind emergency protocols (traffic/industrial curtailment) when stagnation is predicted.

Sector and Business Context

Sector Overview

This project sits at the intersection of **environmental analytics** and **public health policy**. Air quality management is critical for industrialising cities where growth has historically been prioritised over environmental protection. China's NAAQS annual limit is 35 μg/m^3, still well above the WHO's 15 μg/m^3 standard.

Decision-Maker / Stakeholder

- **Beijing Municipal Environmental Bureau:** To determine pollution alert timing and seasonal curtailment.
- **Urban Public Health Officials:** To schedule advisories for elderly and respiratory patients.
- **Traffic Management:** To trigger vehicle restrictions during low-dispersion windows.

Why This Problem Matters

Beijing's PM2.5 frequently exceeded the ”Very Unhealthy” threshold (250 μg/m^3), contributing significantly to cardiovascular and respiratory mortality in China. Without a data-backed alert framework, responses remain reactive.

Problem Statement and Objectives

Formal Problem Definition

Core Question: Which meteorological conditions and temporal patterns best predict hazardous PM2.5 levels in Beijing, and how can these be operationalised into a decisionsupport framework?

Scope

- **In scope:** Hourly PM2.5 data (US Embassy, Beijing); meteorological variables (2010– 2014); Tableau dashboard construction.
- **Out of scope:** Multi-station spatial interpolation; real-time streaming; predictive ML modeling; economic cost modelling.

Success Criteria

Criterion	Target	Status
Dataset Scale	> 5,000 rows	43,824 rows
KPIs Computed	≥ 4 KPIs	6 KPIs
Statistical Tests	≥ 3 methods	7 analyses
Decision Insights	8–12 insights	10 insights
Actionable Recs	3–5	5 recommendations

Data Description

Source and Access

- **Source:** UCI Machine Learning Repository (Liang et al., 2015).
- **Access URL:** https://archive.ics.uci.edu/ml/datasets/Beijing+PM2.5+Data
- **Granularity:** Hourly readings from the US Embassy, Beijing.

Dataset Size and Coverage

- **Dimensions:** 43,824 rows × 13 columns.
- **Temporal Range:** 1 January 2010 to 31 December 2014.

Key Columns

Column	Description	Role

pm2.5	PM2.5 concentration ($\mu g/m^3$)	Target Variable
DEWP/TEMP	Dew Point and Temperature (°C)	Predictors
cbwd	Combined Wind Direction (NW, NE, SE, cv)	Segment
Iws	Cumulated wind speed (m/s)	Predictor
Ir/Is	Cumulated Rain/Snow hours	Precipitation Flag

Data Quality Issues

- **Nulls:** 2,067 rows (4.72%) missing in `pm2.5`.
- **Categorical Noise:** `cbwd = "cv"` (calm/variable) required specific handling as a valid meteorological state rather than missing data.

Cleaning and Transformation

Major Cleaning Steps

1. **Dropping Indices:** Removed the No column as it carried no analytical value.
2. **Datetime Synthesis:** Merged year, month, day, and hour into a single datetime64 column.
3. **Missing Value Imputation:** Applied forward-fill (then backward-fill) to pm2.5 nulls, assuming temporal autocorrelation.
4. **Snake case Normalisation:** Renamed columns (e.g., pm2.5→pm2 5 cleaned).

Assumptions Made

- PM2.5 levels are temporally autocorrelated; hourly readings do not fluctuate wildly between adjacent hours.
- US EPA AQI breakpoints are valid proxies for health risk categorisation in this context.

Output Dataset Description

- **File:** beijing pm25 cleaned.csv
- **Final Features:** 14 (Original 13 - No + datetime + season + aqi category).

KPI Framework

Definitions and Formulae

KPI	Formula	Rationale
Annual Mean	$\bar{x}_{PM} = \frac{1}{n}\sum PM2.5$	Primary compliance metric.
Hazardous Rate	$\frac{\text{Days with Avg>150}}{365} \times 100$	Quantifies acute risk frequency.
WHO Ratio	$\frac{\bar{x}_{PM}}{15}$	Contextualises pollution against safety standards.
AQI Distribution	$\frac{\text{Hours in Category}}{\text{Total Hours}}$	Visualises time-spent-atrisk.

Exploratory Analysis

Major Trends

PM2.5 is heavily right-skewed (Mean: 97.8, Median: 72), driven by extreme spikes up to 994 $\mu g/m^3$. No consistent downward trend was observed between 2010 and 2014.

Segment-Level Insights

- **Diurnal:** Lowest readings occur at 2:00–4:00 PM (high mixing); peaks occur at night.
- **Wind Direction:** NW winds yield 69.8 $\mu g/m^3$ (lowest), while ”cv” yields 124.5 $\mu g/m^3$ (highest).
- **Seasonality:** Winter (109.9) vs Summer (92.1) difference is statistically significant.

Visual Summaries

EDA yielded 11 plots including Monthly Average Time Series, Hourly Heatmaps, and AQI breakdown donut charts.

9. Statistical Analysis

Method Used

Due to the non-normal distribution of PM2.5 values, non-parametric statistical methods were applied to ensure reliable results. Spearman Rank Correlation was used to analyze relationships between PM2.5 and continuous variables such as wind speed and dew point. Kruskal-Wallis Test was used to compare pollution levels across categories like seasons. Mann-Whitney U Test was used to evaluate the effect of rainfall.

Results

Wind speed shows a moderate negative correlation with PM2.5, indicating that higher wind speeds reduce pollution levels. Dew point has a moderate positive correlation, suggesting that higher humidity increases pollution concentration. Seasonal differences are statistically significant, confirming that pollution varies across seasons. Rainfall does not show any significant impact on pollution levels.

Business Interpretation

Wind speed is the most important natural factor for reducing pollution, making it a key variable for planning interventions. Seasonal variation supports the need for targeted winter pollution control strategies. High humidity conditions increase pollution risk, which should be considered in forecasting models. Rainfall is not a reliable mechanism for reducing pollution, so policies should not depend on it.

Dashboard Walkthrough

Dashboard Objective

The dashboard is designed to help decision-makers monitor pollution levels, identify high-risk periods, and understand the environmental factors driving air quality.

Executive View

The executive view provides high-level KPIs such as Annual Mean PM2.5, WHO comparison levels, and the number of hazardous days. It allows quick assessment of overall air quality performance.

Operational View

The operational view focuses on detailed analysis. It includes hour versus month heatmaps to identify peak pollution periods, wind analysis to understand dispersion patterns, and seasonal breakdowns to highlight high-risk timeframes.

Filters and Interactivity

The dashboard includes filters for year, month, hour, AQI category, season, and wind direction. These filters allow users to drill down into specific scenarios. The dashboard follows a structured flow from severity to timing to drivers, enabling both overview and deep analysis.

11. Key Insights (Decision-Oriented)

1. PM2.5 levels consistently exceed WHO safe limits by 6–7 times.
2. No significant improvement trend is observed from 2010 to 2014.
3. Around 21% of total hours fall under hazardous pollution levels.
4. Approximately 1 in every 5 days is classified as hazardous.
5. Winter is the most polluted season due to heating emissions.
6. Pollution levels peak during late-night and early-morning hours.
7. Calm wind conditions result in the highest pollution levels.
8. Northwest winds significantly reduce pollution levels.
9. Wind speed is the strongest predictor of pollution reduction.
10. Rainfall does not significantly impact pollution levels.
11. Snow-related pollution is seasonal, not causal.
12. Summer months show relatively lower pollution levels.

Recommendations

- **Recommendation 1:** Launch Winter-Priority Emission Reduction targeting coal heating (Dec-Feb).
- **Recommendation 2:** Deploy NW-Wind ”Clean Window” alerts when NW winds exceed 21 m/s.
- **Recommendation 3:** Implement Calm-Wind Emergency Protocols (cbwd=cv) including industrial caps.
- **Recommendation 4:** Shift to Hourly AQI monitoring for health advisories to capture diurnal spikes.
- **Impact:** Potential 10–15% reduction in peak exposure and improved public health response timing.

Limitations and Next Steps

Limitations

- **Spatial:** Data represents a single monitoring station (US Embassy).
- **Policy:** Dataset predates major 2013 Action Plan interventions.
- **Causality:** No source apportionment; drivers are meteorological proxies only.

Future Work

- Multi-station analysis to map spatial pollution gradients across Beijing.
- Development of SARIMA or XGBoost models for predictive forecasting.
- Correlation of air quality data with hospital admission records.

Name	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Milind B.	Owner	Support	—	Support	Support	—
Hrishabh	—	Support	Support	Owner	—	—
Om	Support	—	—	—	Owner	Owner
Vishuti J.	—	Owner	Owner	—	Support	—
Samay S	—	—	Support	—	Owner	Support
Praveen N.	—	Support	Support	Owner	—	—